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PLB 111 — Past Questions & Materials

by Renance

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Paper 1 of 1 — PLB111 PQ

Level 100 • First Semester • Handout

PLB 111: PLANT DIVERSITY (11th December, 2021)

GROWTH AND DEVELOPMENT IN HIGHER PLANTS

Have you ever thought about where and how the structures like roots, stems, leaves, flowers, fruits and

seeds arise and that too in an orderly sequence? You are, by now, aware of the terms seed, seedling,

plantlet, mature plant. You have also seen that trees continue to increase in height or girth over a period

of time. However, the leaves, flowers and fruits of the same tree not only have limited dimensions but

also appear and fall periodically and some time repeatedly.

Why does vegetative phase precede flowering in a plant? All plant organs are made up of a variety of

tissues; is there any relationship between the structure of a cell, a tissue, an organ and the function they

perform? Can the structure and the function of these be altered? All cells of a plant are descendents of

the zygote. The question is, then, why and how do they have different structural and functional attributes?

Plant Development is the sum of two processes: growth and differentiation. To begin with, it is essential

and sufficient to know that the development of a mature plant from a zygote (fertilised egg) follow a

precise and highly ordered succession of events. During this process a complex body organisation is

formed that produces roots, leaves, branches, flowers, fruits, and seeds, and eventually they die (Figure

1). The first step in the process of plant growth is seed germination. The seed germinates when favourable conditions for growth exist in the environment. In absence of such favourable conditions the

seeds do not germinate and goes into a period of suspended growth or rest. Once favourable conditions

return, the seeds resume metabolic activities and growth takes place.

In this chapter, you shall also study some of the factors which govern and control these developmental

processes. These factors are both intrinsic (internal) and extrinsic (external) to the plant.

Figure1: Germination and seedling development in bean

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GROWTH

Growth is regarded as one of the most fundamental and conspicuous characteristics of a living being.

What is growth? Growth can be defined as an irreversible permanent increase in size of an organ or its

parts or even of an individual cell. Generally, growth is accompanied by metabolic processes (both anabolic and catabolic), that occur at the expense of energy. Therefore, for example, expansion of a leaf

is growth. How would you describe the swelling of piece of wood when placed in water?

Plant Growth generally is Indeterminate

Plant growth is unique because plants retain the capacity for unlimited growth throughout their life. This

ability of the plants is due to the presence of meristems at certain locations in their body. The cells of

such meristems have the capacity to divide and self-perpetuate. The product, however, soon loses the

capacity to divide and such cells make up the plant body. This form of growth wherein new cells are always being added to the plant body by the activity of the meristem is called the open form of growth.

What would happen if the meristem ceases to divide? Does this ever happen? In earlier discussion, you

have studied about the root apical meristem and the shoot apical meristem. You know that they are responsible for the primary growth of the plants and principally contribute to the elongation of the plants

along their axis. You also know that in dicotyledonous plants and gymnosperms, the lateral meristems,

vascular cambium and cork-cambium appear later in life. These are the meristems that cause the increase in the girth of the organs in which they are active. This is known as secondary growth of the plant (Figure 2).

Figure 2: Diagrammatic representation of locations of root apical meristem, shoot apical meristem and

vascular cambium. Arrows exhibit the direction of growth of cells and organ

Growth is Measurable

Growth, at a cellular level, is principally a consequence of increase in the amount of protoplasm. Since

increase in protoplasm is difficult to measure directly, one generally measures some quantity which is

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more or less proportional to it. Growth is, therefore, measured by a variety of parameters some of which

are: increase in fresh weight, dry weight, length, area, volume and cell number. You may find it amazing

to know that one single maize root apical meristem can give rise to more than 17,500 new cells per hour,

whereas cells in a watermelon may increase in size by up to 3,50,000 times. In the former, growth is

expressed as increase in cell number; the latter expresses growth as increase in size of the cell. While the

growth of a pollen tube is measured in terms of its length, an increase in surface area denotes the growth

in a dorsiventral leaf.

Phases of Growth

The period of growth is generally divided into three phases, namely, meristematic, elongation and maturation (Figure 3). Let us understand this by looking at the root tips. The constantly dividing cells,

both at the root apex and the shoot apex, represent the meristematic phase of growth. The cells in this

region are rich in protoplasm, possess large conspicuous nuclei. Their cell walls are primary in nature,

thin and cellulosic with abundant plasmodesmatal connections. The cells proximal (just next, away from

the tip) to the meristematic zone represent the phase of elongation. Increased vacuolation, cell enlargement and new cell wall deposition are the characteristics of the cells in this phase. Further away

from the apex, i.e., more proximal to the phase of elongation, lies the portion of axis which is undergoing the phase of maturation. The cells of this zone, attain their maximal size in terms of wall

thickening and protoplasmic modifications. Most of the tissues and cell types you have studied in earlier chapter represent this phase.

Fig. 3: Detection of zones of elongation by the parallel line technique. Zones A, B, C, D immediately behind the apex have elongated most

Growth Rates

The increased growth per unit time is termed as growth rate. Thus, rate of growth can be expressed mathematically. An organism or a part of the organism can produce more cells in a variety of ways. The

growth rate shows an increase that may be arithmetic or geometrical (4). In arithmetic growth, following

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mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. The simplest expression of arithmetic growth is exemplified by a root elongating at a constant

rate. Look at Figure 5. On plotting the length of the organ against time, a linear curve is obtained. Mathematically, it is expressed as

$$L_t = L_0 + rt$$

L_t = length at time 't'

L_0 = length at time 'zero'

r = growth rate / elongation per unit time.

Figure 5: Constant linear growth, a plot of length L against time t

Let us now see what happens in geometrical growth. In most systems, the initial growth is slow (lag phase), and it increases rapidly thereafter – at an exponential rate (log or exponential phase). Here, both

the progeny cells following mitotic cell division retain the ability to divide and continue to do so.

However, with limited nutrient supply, the growth slows down leading to a stationary phase. If we plot

the parameter of growth against time, we get a typical sigmoid or S-curve (Fig. 6). A sigmoid curve is a

characteristic of living organism growing in a natural environment. It is typical for all cells, tissues and

organs of a plant. Can you think of more similar examples? What kind of a curve can you expect in a

tree showing seasonal activities?

The exponential growth can be expressed as follows:

$$W_1 = W_0 e^{rt}$$

W_1 = final size (weight, height, number etc.)

W_0 = initial size at the beginning of the period

r = growth rate

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t = time of growth

e = base of natural logarithms

Here, r is the relative growth rate and is also the measure of the ability of the plant to produce new plant

material, referred to as efficiency index. Hence, the final size of W_1 depends on the initial size, W_0 .

Figure 6: An idealised sigmoid growth curve typical of cells in culture, and many higher plants and plant

organs

Conditions for Growth

Why do you not try to write down what you think are necessary conditions for growth? This list may have moisture/water, oxygen and nutrients as very essential elements for growth. The plant cells grow in

size by cell enlargement which in turn requires water. Turgidity of cells helps in extension growth. Thus,

plant growth and further development is intimately linked to the water status of the plant. Water also

provides the medium for enzymatic activities needed for growth. Oxygen helps in releasing metabolic

energy essential for growth activities. Nutrients (macro and micro essential elements) are required by

plants for the synthesis of protoplasm and act as source of energy.

In addition, every plant organism has an optimum temperature range best suited for its growth.

Any

deviation from this range could be detrimental to its survival. Environmental signals such as light and gravity also affect certain phases/stages of growth.

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DIFFERENTIATION, DE-DIFFERENTIATION AND RE-DIFFERENTIATION

The cells derived from root apical and shoot-apical meristems and cambium differentiate and mature to

perform specific functions. This act leading to maturation is termed as differentiation. During differentiation, cells undergo few to major structural changes both in their cell walls and protoplasm. For

example, to form a tracheary element, the cells would lose their protoplasm. They also develop very

strong, elastic, lignocellulosic secondary cell walls, to carry water to long distances even under extreme

tension. Try to correlate the various anatomical features you encounter in plants to the functions they

perform. Plants show another interesting phenomenon. The living differentiated cells that by now have

lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon

is termed as dedifferentiation. For example, formation of meristems – interfascicular cambium and cork

cambium from fully differentiated parenchyma cells. While doing so, such meristems/tissues are able to

divide and produce cells that once again lose the capacity to divide but mature to perform specific functions, i.e., get re-differentiated. List some of the tissues in a woody dicotyledonous plant that are the

products of re-differentiation. How would you describe a tumour? What would you call the parenchyma

cells that are made to divide under controlled laboratory conditions during plant tissue culture?

Recall,

we have mentioned that the growth in plants is open, i.e., it can be indeterminate or determinate.

Now,

we may say that even differentiation in plants is open, because cells/tissues arising out of the same meristem have different structures at maturity. The final structure at maturity of a cell/tissue is also determined by the location of the cell within. For example, cells positioned away from root apical meristems differentiate as root-cap cells, while those pushed to the periphery mature as epidermis.

Can

you add a few more examples of open differentiation correlating the position of a cell to its position in

an organ?

DEVELOPMENT

Development is a term that includes all changes that an organism goes through during its life cycle from

germination of the seed to senescence. Plants follow different pathways in response to environment or

phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in

cotton, coriander and larkspur. In such plants, the leaves of the juvenile plant are different in shape from

those in mature plants. On the other hand, difference in shapes of leaves produced in air and those produced in water in buttercup also represent the heterophyllous development due to environment. This

phenomenon of heterophylly is an example of plasticity.

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Characteristics

The plant growth regulators (PGRs) are small, simple molecules of diverse chemical composition. They

could be indole compounds (indole-3-acetic acid, IAA); adenine derivatives (N6-furfurylamino purine,

kinetin), derivatives of carotenoids (abscisic acid, ABA); terpenes (gibberellic acid, GA3) or gases (ethylene, C₂H₄). Plant growth regulators are variously described as plant growth substances, plant hormones or phytohormones in literature.

The PGRs can be broadly divided into two groups based on their functions in a living plant body. One

group of PGRs are involved in growth promoting activities, such as cell division, cell enlargement, pattern formation, tropic growth, flowering, fruiting and seed formation. These are also called plant growth promoters, e.g., auxins, gibberellins and cytokinins. The PGRs of the other group play an important role in plant responses to wounds and stresses of biotic and abiotic origin. They are also involved in various growth inhibiting activities such as dormancy and abscission. The PGR abscisic acid

belongs to this group. The gaseous PGR, ethylene, could fit either of the groups, but it is largely an inhibitor of growth activities.

Auxin - It is commonly known as indole-3-acetic acid (IAA). It is produced at the stem and root apex and often migrates to the site of action.

It serves the following functions:

1. Cell enlargement.
2. Cell division.
3. Apical dominance.
4. Induce Parthenocarpy.
5. Inhibition of abscission.

Gibberellins: Gibberellins are promotory PGR seen in more than 100 forms named as They are denoted

as GA1, GA2, GA3 and so on. Gibberellic Acid is the most common one.

It serves the following functions:

1. Cell elongation.

2. Early maturity.
3. Seed germination.
4. Breaking of dormancy.

Cytokinins - Cytokinins have specific effects on cytokinesis and were discovered as kinetin (a modified

form of adenine, a purine) from the autoclaved herring sperm DNA. The most common forms are zeatin,

kinetin, etc. They are mainly made in the roots.

Some of the Functions:

1. Cell division and cell differentiation.
2. Overcome apical dominance.
3. Promote nutrient mobilization.
4. Essential for tissue culture.

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Ethylene - It is called a gaseous hormone that stimulates isodiametric or transverse growth; however, it

retards the longitudinal one.

It serves the following functions:

1. Inhibition of longitudinal Growth.
2. Senescence.
3. Promote apical dominance.
4. Fruit ripening.

Abscisic Acid - It is also referred to as stress hormone or dormin. It works like a general plant growth

inhibitor. Absciscic acid is formed at the terminal buds of the top of the plant or in the roots of the plants.

It can serve the following functions:

1. Bud dormancy.
2. Induce Parthenocarpy.
3. Seed development and maturation.
4. Leaf senescence.

Photoperiodism - Photoperiodism is the effect of photoperiods or day duration of light hours on the plant's growth and development, especially flowering. The flowering plants have been divided into the

following categories based on photoperiodic response:

1. Short Day Plants - The flowers which need exposure to light for a period less than this critical duration before the flowering is initiated. For example - Xanthium, Sugarcane, Potato Rice.
2. Long Day Plants - The plant flowers when they need a long photoperiod of light, greater than the critical period. Example - Barley, Radish, Lettuce.
3. Day Neutral Plants - These plants can blossom throughout the year - for example - Wild Kidney, Bean.

Vernalisation - is the process of reducing the juvenile or vegetative phase and fastening the flowering

procedure by cold treatment. Meristematic cells help in perceiving the stimulus of vernalisation.

Vernalisation reduces the vegetative period of plants and leads to early flowering. It applies to temperate

plants like Rice, Wheat, Millets, etc.

Summary of Processes of Growth and Development in Higher Plants

- The stem, roots, leaves, flowers, fruits, and seeds of the plants arise naturally. The order of the growth is in the following manner:
- The plants begin their reproductive phase, where the flowers and fruits are produced to continue the plants' life cycle after completing their vegetative phase.
- Development is the sum of two processes which is growth and differentiation. Few internal and external factors control development and growth.
- Growth is an irreversible increase in dry weight, size, mass, or volume of cell, organ, or organism which is permanent. It is usually internal in living organisms.
- Growth is achieved by cell division, which increases cell number and cell enlargement in the plants. Hence, growth is a quantitative aspect which can be measured according to time.
- Plant growth is generally indeterminate because of the capacity of unlimited growth in their lifespan. Meristem tissues are there at certain parts of the plant body.

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- The Open Form of Growth - The plant growth in which there is an addition of new cells to the plant body due to meristem.
- Root Apical Meristem and Shoot Apical - Meristem leads to elongation and the primary growth of the plant body along with the axis.
- At the cellular level, growth refers to the increase in the amount of protoplasm. It is often difficult to measure this increase. However, we can measure the increase in the cell, cell number, and cell size.
- The growth measure is checked by the increase in fresh weight, length, dry weight, area, volume, and cell number. Some can be measured for measuring some kinds of growth.
- The phase of cell division or cell formation is called the formative phase. It takes place at the shoot apex, root apex, and other regions having meristematic tissue. The rate of respiration is very usually high in the cells undergoing mitosis division in the formative phase.
- Enlargement Phase - The newly formed cells produced in the formative phase will undergo enlargement. This enlargement leads to the development of vacuoles that further lead to an increase in cells' volume.
- Cell enlargement takes place from all directions with maximum elongation in conducting tissues and fibres.
- Water, oxygen, and essential elements are the essential conditions for growth. Water is highly required for all the cell enlargement and controls the turgidity. Water also acts as a medium for enzymatic conditions.
- Water is required for the formation of Protoplasm, while the micro and macronutrients act as a

source of energy.

- Plants also need optimal temperature and other environmental conditions.
- The act of maturation where the cells formed by apical meristem turn into a specialized one to serve a specific function is called differentiation.
- The differentiated living cells which are living that no longer can undergo division can regain the capacity of it. This phenomenon is known as dedifferentiation, for instance, cork cambium and interfascicular cambium and cork cambium.
- Re-differentiation is a process where the dedifferentiated cells mature and lose cell division capacity again to serve specific functions.

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